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# Multiple Outputs Converter Design for BMS Integrated Li-Ion Battery Charger Appropriate for Electric Vehicle

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**Abstract-** This paper presents a high-power density onboard integrated charger with multiple outputs-based H-bridge converters for BMS (Battery Management System) integrated li-ion battery bank used in upgraded electric vehicles. The converter is proposed to function as AC/DC battery charger and to transfer energy from the supply grid to multiple battery packs through discrete BMS of the system drivetrain. The BMS controls over voltage, overcurrent, and temperature of the battery pack. This operation will be done more efficiently with the separate power supply to these discrete BMSs. To obtain a high-power density and high efficiency, an HF tapped-transformer is used to meet the requirement of improved system size and reliability by attenuating magnetizing and copper losses with high-switching frequency operation. But the conventional H-bridge circuit suffers from switching loss due to HF operation. To solve this, zero-voltage switching (ZVS) topology was integrated. The impact of the proposed topology and control strategy on the integrated charger power losses, efficiency, and power density has been simulated and analyzed by computer simulation and practical implementation.

**Index Terms-** charger, electric vehicle, li-ion, BMS (Battery Management System), HF transformer, high-efficiency.

## I. INTRODUCTION

Electric vehicles (EVs) or easy bikes are the most used transportation in south Asia. Due to cost-effectiveness, easy maintenance with zero carbon emission and no requirement of gas encourages people to use electric vehicles. But this EV mainly runs by the power backup from bulky lead-acid batteries. Five 12V batteries, 100AH needs to be fully charged daily such that a vehicle like an e-bike for at least 5 hours. [1]. The battery bank weight 150kg and 3 kWh of energy are required to fully charge such a battery bank.

Alternatively, to charge this battery bank, using chargers make if bulky off-board iron core transformer with a capacity less than 75% [2]. The basic working principle of this type of charger is that the transformer receives input from the utility grid and then lowers the voltage through the taps. The output charging current is controlled by a manual switch which is linked with the tapped wires. To create usable DC power for charging the battery bank, the output of the transformer was rectified. The charging current varies from 5 A to 20 A depending on the battery voltage. Therefore, the wattage required to charge the battery bank varies from 300 W to 1200 W. The transformer experiences power losses like hysteresis loss, eddy current loss, resistive losses in the copper coil, and magnetizing loss in the core during this power transition and resulting in more than 30% power loss.

The typical battery bank and charger are bulky, not portable, and inefficient. Therefore, it is highly desirable to have a system with lightweight and efficient. Updated EV systems like “Tesla” (Fig.1), are using lithium-ion batteries

for higher power density and power electronics-based chargers able to operate with more than 90% efficiency which can make the overall system more efficient [4].

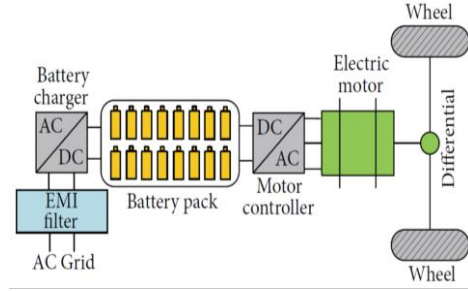


Fig. 1. The block-diagram of the onboard electric vehicle charging system

“Li-ion battery technology has proven to have a higher energy density than that of lead-acid batteries. Higher energy density means it can store more energy in the same physical space, which also means higher capacity. A higher depth of discharge is also possible with lithium-ion technology. It can be discharged 85% where lead-acid batteries should not be discharged more than 50%.” [7] “Most lithium-ion batteries are 95% efficient or more. On The Other Hand, lead-acid batteries see efficiencies closer to 80% to 85%. Li-ion batteries usually last several times the number of cycles as lead-acid batteries, resulting in a longer service life of lithium-ion products.” [5]

Mainly, HF onboard portable charger system used to charge li-ion battery bank which is compact and lightweight. But to maintain high charging and discharging current from the li-ion battery bank, temperature, individual cell voltage, the evenly distributed current should be monitored thoroughly [9]. To make high capacity, a separated bank of the cell should be assembled. These banks can be charged from the same charger with individual inputs. The faster battery monitoring system can be optimized by using specific BMS for these battery banks [10].

To make an efficient li-ion charger, a full-bridge converter with multiple output-based topologies is proposed. After rectification of the grid voltage, the dc/dc converter will be used to control and step down the input voltage to 50Vdc in each output branch by increasing the charging current maximum to 10A. The 1kW prototype inverter was designed with the help of simulation software (PSIM and PROTEUS) and the performance of the proposed system was verified by field testing. The main test results are presented in this article for the onboard charger for BMS integrated li-ion battery bank.

## II. MULTIPLE OUTPUTS CONVERTER TOPOLOGY

In this proposed system, the charger will take the ac voltage from the grid. A full-wave diode rectifier will be used to convert this ac voltage to dc. The rectified output is fed to the H-bridge converter. Fig.2 represents the schematic of the system topology of the multi-output converter-based li-ion charger.

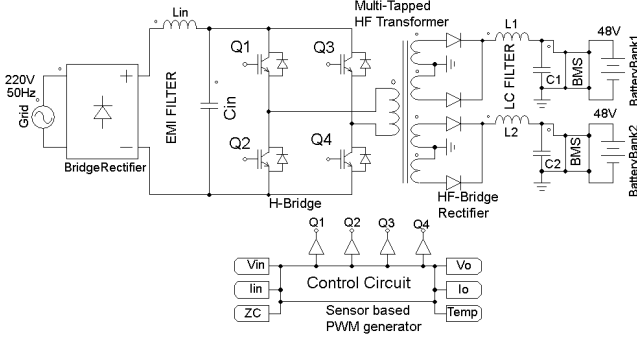


Fig. 2. System topology of the multi-output converter based charger

High-Frequency (HF) power converters nowadays find a great deal of application for electric vehicle (EV) battery charging in either direct plugin or contactless power transfer-based strategies. For this portable onboard charger, 220V 50Hz grid ac supply is considered. The operation of the system can be divided into the following subsections: (A) diode rectifiers, (B) Multi tapped high-frequency transformer, (C) Individual Outputs. The Grid AC is rectified and filtered with the combination of inductors and capacitors. The creation of an AC link across the transformer windings is achieved by switching Q1, Q2, Q3, Q4 which operates with simple pulse width modulation (PWM). For the PWM control strategy, a sinusoidal waveform as the reference signal is generated to do the comparison with the triangular waveform to generate the pulsating gate signal. The reference signal consists of the parameters such as the ac mains voltage and the output current amplitude.

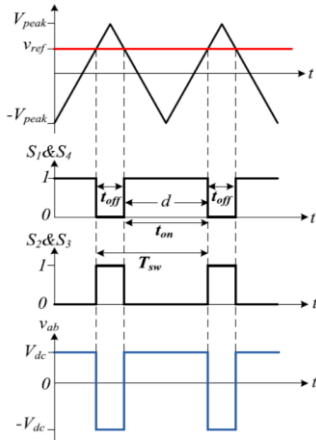


Fig. 3. PWM generation from sensor and modulation signals.

Compensation for sinusoidal,  $V_{ref}$ , and triangular carrier,  $V_{tri}$  waveform is compared producing PWM gateway switching power signals. The main waveforms of a typical PWM design for the two switching cycles are shown in Fig. 3. The  $V_{ref}$  reference signal can be displayed as a straight red line because the switching frequency is much higher than the frequency of the line [11]. The duty ratio  $d$  of the gate signal determines the complementary turn-on time,  $t_{on}$ , and turn-off time,  $t_{off}$ , for the switches pair, Q1 & Q4, and Q2 & Q3,

respectively. The voltage  $V_{ab}$  across terminals A and B are either  $+V_{dc}$  or  $-V_{dc}$ . By modulating  $d$ , the voltage  $V_{ab}$  can be controlled. By referring to the waveforms, the duty ratio  $d$  of an inverter is

$$d = \frac{1}{2} + \frac{v_{ref}}{2V_{peak}} \quad (1)$$

Where  $V_{peak}$  is the highest value of the triangular waveform,  $V_{tri}$ . An equal amount of  $V_{ab}$  within a single switching period is expressed as

$$v_{ab} = [V_{dc} * d] + [-V_{dc} * (1-d)] = [2d-1] V_{dc} \quad (2)$$

By inserting (1) into (2), the voltage indicator in circuits A and B becomes

$$v_{ab} = \left( \frac{V_{dc}}{V_{peak}} \right) * v_{ref} \quad (3)$$

Q1, Q4 switch function has a  $180^\circ$  phase shift compared to the Q2, Q3 switch function. The gates of Q1, Q4, and the gates of Q2, Q3 switch alternatively with a phase shift of  $180^\circ$  to produce a sine wave. This is then fed into the primary side of the transformer. The transformer produces two separate output filters with a single combination of LC filters to achieve stable DC output.

### III. PROTOTYPE CHARGER AND HF TRANSFORMER DEVELOPMENT

According to the simulation, an inverter is made (Figure 4), and the parts used in the construction are placed in Table I and described in the side sections.



Fig. 4. Topside and bottom side of the designed prototype charger.

TABLE I. COMPONENTS USED IN THE EXPERIMENTAL PROTOTYPES.

| Components      | Type                                                                                                            |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Capacitors      | 6 polyester film capacitors of value 0.05 uF, one 330uF 600V Electrolyte Capacitor.                             |
| Inductors       | 5mH, 22 turns of Litz wire wound coil, Ferrite Core                                                             |
| MOSFET          | 2SK3451; $V_{DS} = 600V$ , $R_{DS(ON)} = 0.65 \Omega$ , $V_{GS} = \pm 30V$ , $I_D = 13A$ .                      |
| switch          |                                                                                                                 |
| Microcontroller | PIC18F4431; 40-Pin Nano Watt Technology Microcontroller with High-Performance PWM, and A/D                      |
| Driver          | IR2104; high-speed, high voltage power MOSFET and IGBT drivers with dependent high and low side                 |
| Power Diodes    | 600V, 30A ultrafast diode; MUR3060PT                                                                            |
| Diodes          | STTH102C; 200V, 1A high efficiency fast diode                                                                   |
| Signal Diodes   | 1N4154, Small Signal Fast Switching Diodes                                                                      |
| Regulator       | 5V regulator-LM7805. 15V regulator-LM7815                                                                       |
| Resistors       | 1/4 watt resistors. Two 7-Watts ceramic resistors and two 2-Watts ceramic resistors.                            |
| Temp Sensor     | LM35, Precision Centigrade Temperature Sensor                                                                   |
| Op-Amp          | LM358-N, Two Internally Compensated Op Amps, Compatible with All Forms of Logic.                                |
| Photocoupler    | PC817A, a 4pin DIP, isolation voltage(rms) is 5.0kV, Current transfer ratio (CTR : MIN. 50% at IF=5 mA, VCE=5V) |

The rectified DC output is fed to the H-bridge inverter. In this region the switching transistor is the power MOSFET, both IGBT and MOSFETs are used in power switching, choice is determined by the current, voltage, switching speed,

and cost. After the EMI filter, the rectified DC (input) ripple voltage is smoothened by 600V, 330uF electrolyte capacitor used as  $C_{in}$ . Mica capacitor  $C_{rs}$  of value 560 pF was used to provide the correct net capacitance. This inverter contains mainly four switches. The 30kHz PWM signal is generated using a microcontroller (MCU). The MCU detects the input current and voltage such that the reference input voltage can control the duty cycle of the PWM signal. The MOSFET driver (IR2104) is used to drive high capacity power MOSFET (2SK3451). A temperature sensor (LM35) is used to keep heat in check. The control circuit derives its power from LM7815 and LM7805 voltage regulators. To sense the input voltage, a PC817A photocoupler was used that can provide isolation for the control circuit from HV. Two 0.05 ohm, 5W resistance was placed in parallel in the negative line. The voltage differences between these registers and the ground reference are taken to measure the flow of current. A combination of full-wave rectifier and comparator (op-amp LM358N) are used to detect zero crosses. The modulation index reference signal is also taken from this part of the sensing circuitry. The operational specifications of the charger are shown in Table II.

TABLE II. PARAMETERS SPECIFICATIONS OF THE CHARGER

| Parameters             | Values              |
|------------------------|---------------------|
| System Capacity        | 1000 W              |
| Switching Frequency    | 30 kHz              |
| Efficiency             | < 93 %              |
| Input DC Voltage       | 310 V <sub>DC</sub> |
| Output Voltage         | 48 V <sub>DC</sub>  |
| Maximum Output Current | 10 A                |

The inverter output is an HF AC fed to the HF tapped-transformer (22: 6-0-6). Mainly two output has been taken from this transformer to charge two different li-ion battery pack. The transformer core size can determine by reference [12] that the design of an HF transformer can be summarized in the following equations.

$$K_{gfe} \geq \frac{\rho \lambda_1^2 I_{tot}^2 K_{fe}^{(2/\beta)}}{4K_{\mu}(P_{tot})^{((\beta+2)/\beta)}} * 10^8 \quad (4)$$

Here,

$\rho$ , Wire effective resistivity ( $\Omega$ -cm), for copper,  $1.724*10^{-6}$ .

$I_{tot}$ , Total RMS winding current, ref to primary (A)

$\lambda_1$ , Applied primary volt-sec (Volt-sec),  $\lambda_1 = V_p.T_s$ . D; D, Duty Cycle,

$V_p$ , Primary Voltage (average during ON period);  $T_s = 1/f$

$P_{tot}$ , Allowed total power dissipation.

$K_{\mu}$ , Winding fill factor (0.3 – 0.5), Typical 0.4

$B$ , Core loss exponent, Typical = 2.6

$K_{fe}$ , Core loss coefficient ( $W/cm^3 T^\beta$ ), 2.5  $W/cm^3 T^\beta$

From the database in [13], by calculating  $K_{gfe}$ , the relevant core types for this application, ETD29 is used that can hold the required copper coil. It is possible to reduce the core size by selecting a core that has low losses. A higher magnetic permeability means, a lower amount of eddy current and lower switching losses.

$A_c$ , Core cross-sectional area = 4.2  $cm^2$

$W_A$ , Core window area = 2.5  $cm^2$

MLT, Mean length per turn = 12.5 cm

$l_e$ , Magnetic path length = 12.4 cm

$A_{w1}$ , Wire area = 0.00326  $cm^2$

$\Delta B$ , Peak ac flux density = 0.08 T

A magnetic material normally tends to generate an eddy current in response to a rising magnetic frequency. Since higher frequencies allow a proportionately lower number of turns, results in the transformer being smaller, lighter, and

cheaper. This is why SMPS uses a high frequency. The design parameters of the transformer are given in Table III.

TABLE III. PARAMETERS SPECIFICATIONS OF THE TRANSFORMER

| Step-down Transformer | Primary  | Secondary 1      |       | Secondary 2       |       |
|-----------------------|----------|------------------|-------|-------------------|-------|
| Windings              | 24 turns | 6-0-6            |       | 6-0-6             |       |
| Resistance            | 0.3 ohm  | 0.1 ohm          |       | 0.1 ohm           |       |
| Inductance            | 3.18mH   | 0.2mH            | 0.2mH | 0.19mH            | 0.2mH |
|                       |          | 0.8mH (combined) |       | 0.83mH (combined) |       |

The center-tapped transformer primary coil was completed by 22 turns using 6 numbers of 20 SWG super enameled copper wire, in parallel. The specification of the transformer is determined by the application in which the transformer is used. The following information in Table IV is the design specification for an H-bridge transformer, operating at 30 kHz, using the  $K_{gfe}$  core geometry approach.

TABLE IV. DESIGN PARAMETERS OF THE TRANSFORMER

| Parameter                   | Symbol                      | Value& Unit                            |
|-----------------------------|-----------------------------|----------------------------------------|
| Input Voltage               | $V_p$                       | 220V <sub>ac</sub> /300V <sub>dc</sub> |
| Output Voltage #1           | $V_{O1}$                    | 65V <sub>dc</sub>                      |
| Output Current #1           | $I_{O1}$                    | 5A                                     |
| Output Voltage #2           | $V_{O2}$                    | 65V <sub>dc</sub>                      |
| Output Current #2           | $I_{O2}$                    | 5A                                     |
| Frequency                   | $f$                         | 30kHz                                  |
| Efficiency                  | $\eta$                      | 95%                                    |
| Regulation                  | $\alpha$                    | 0.5%                                   |
| Diode voltage drop          | $V_d$                       | 0.7V                                   |
| Operating Flux Density      | $B_{ac}$                    | 0.1T                                   |
| Window Utilization          | $K_u$                       | 0.4                                    |
| Temperature rise            | $T_r$                       | 30°C                                   |
| Summation of Output Current | $\sum_{i=1}^n I_{oi} - max$ | 10A                                    |

The two outputs are rectified and then supplied to the LC filter to remove any DC voltage ripple before it is applied to the BMS. Litz wire was selected for winding inductors L. Electrolyte capacitor bank  $C_O$  of value 220uF 200V was used. The parameters of the LC filter design are given in Table V.

TABLE V. PARAMETERS SPECIFICATIONS OF THE LC FILTER

| LC filter | Output 1          | Output 2          |
|-----------|-------------------|-------------------|
| Inductor  | 3.17mH (22 turns) | 3.16mH (23 turns) |
| Capacitor | 220uF/200V        | 220uF/200V        |

The ferrite section is constructed around a 30mm toroid core. Concerning the architecture, the L1 inductor has 50turns 22SWG copper wire. The same procedure follows for L2.

#### IV. SYSTEM TESTING AND RESULT ANALYSIS

The circuit was assembled according to the design and carried out experimentation to justify the performance in the laboratory. For experimentation, a 5kVA ac VARIAC was used to supply different input voltages from the grid. The ac voltage from the grid is first converted to dc using a full-wave diode rectifier. The rectified output is fed to the full-bridge inverter and the output of the inverter is high-frequency ac voltage which is applied across the ferrite core transformer to step down the voltage as required for charging operation. However, voltage is ac and therefore must be converted to dc using a multi-tap rectifier. Finally, an LC filter is used to remove any ripple from the dc voltage before it is applied across the BMS of the battery bank for charging. Fig.5 shows



the entire arrangement of the proposed charger in Lab. The charging current was varied using the VARIAC, and the corresponding input/output voltage and current were recorded using analog meters.

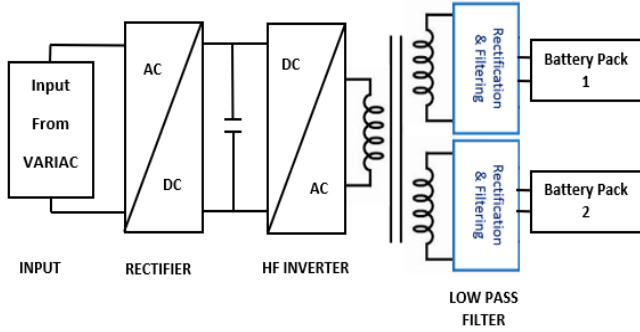


Fig 5. Block diagram of experimentation with the designed prototypes.

The oscilloscope was used to observe the waveforms generated from the inverter. Fig.6 shows the two output waveforms of the H-bridge inverter and final dc output.

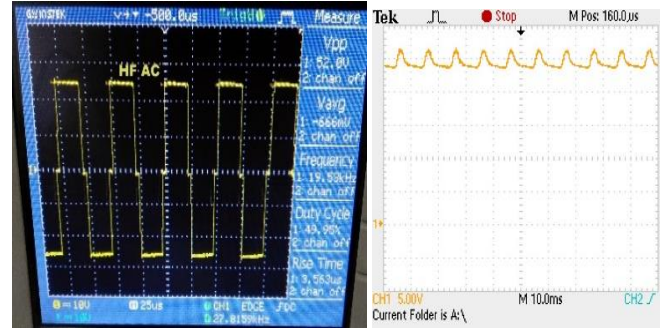


Fig 6. The output of the system from the Oscilloscope. a) Inverter output waveforms (HF AC) and b) DC/DC power converter output waveforms showing DC voltage with ripple (yellow).

The output power was measured by calculating the product of the charging current and the voltage that appeared across the battery bank. Finally, using the ratio of output to input power, the efficiency of the charging system was calculated. Table VI below shows the experimental data obtained from the system.

TABLE VI. PARAMETERS SPECIFICATIONS OF THE CHARGE CONTROLLER

| $V_{in}$ (V) | $I_{in}$ (A) | $P_{in}$ (W) | $V_o$ (V): A | $I_o$ (A): A | $P_o$ (W): A | $V_o$ (V): B | $I_o$ (A): B | $P_o$ (W): B | Total $P_o$ (W) | $\eta$ (%) |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-----------------|------------|
| 210          | 0.07         | 14.7         | 48.5         | 0.10         | 4.850        | 49.8         | 0.11         | 5.48         | 10.33           | 70.26      |
| 220          | 0.07         | 15.4         | 49.0         | 0.10         | 4.900        | 49.9         | 0.11         | 5.49         | 10.39           | 67.46      |
| 230          | 0.14         | 32.2         | 50.0         | 0.20         | 10.00        | 50.0         | 0.21         | 10.50        | 20.50           | 63.66      |
| 240          | 0.80         | 192          | 50.2         | 1.30         | 65.26        | 51.0         | 1.30         | 66.30        | 131.56          | 68.52      |
| 250          | 2.00         | 500          | 52.3         | 3.66         | 191.42       | 52.0         | 3.65         | 189.80       | 381.22          | 76.24      |
| 260          | 2.00         | 520          | 54.3         | 4.20         | 228.06       | 54.5         | 4.50         | 245.25       | 473.31          | 91.02      |
| 270          | 2.65         | 716          | 56.3         | 5.80         | 326.54       | 56.0         | 6.00         | 336.00       | 662.54          | 92.60      |
| 280          | 3.40         | 952          | 57.7         | 7.60         | 438.52       | 57.4         | 7.50         | 430.50       | 869.02          | 91.28      |
| 290          | 3.70         | 1073         | 59.0         | 8.00         | 472.00       | 60.0         | 8.40         | 504.00       | 976.00          | 90.96      |
| 300          | 4.12         | 1236         | 61.0         | 9.00         | 549.00       | 62.0         | 9.20         | 570.40       | 1119.40         | 90.57      |

The parameters of the charge controller have been tabled above. The minimum working voltage for the controller is  $200V_{DC}$ . The input voltage range is taken to be between 200 to 300 DC volts, with 311 volts to be the max DC voltage (considering grid input  $220V_{AC}$  volt). There are two separate isolated center-tapped outputs A and B. Individual Current-Voltage output readings are obtained, the power of output A, output B, and total Power is calculated. At  $210V_{DC}$ , the output voltage is around 48.5 to 50 volts with an average of 0.1A, with a total power of 10W. With  $300V_{DC}$ , the desired output voltage is obtained at  $62 V_{DC}$  ( $\pm 2V_{DC}$ ),  $9A$  ( $\pm 0.2A$ ) current. The current draw at  $300V_{DC}$  is 4.12A resulting in total power output of 1120W. The efficiency of the system is 90%.

A simulation of the topology was carried out in Proteus Simulation software such as in database [1], the theoretical efficiency observed was around 95%. Upon practical implementation, the efficiency achieved was around 90%. External factors such as heat loss, non-linear characteristics of used components played a major role in reduced efficiency.

In Fig 7, the y-axis represents the branch current (A), the x-axis on the left represents the power (W) consumption while the x-axis on the right represents the efficiency against branch current (A). The blue line represents output current, the grey line represents efficiency while the orange portion represents output power (W) by the system. It should be

mentioned that the demanded output power command is set by the user and the dc input voltage remains constant during the hardware experimental test.

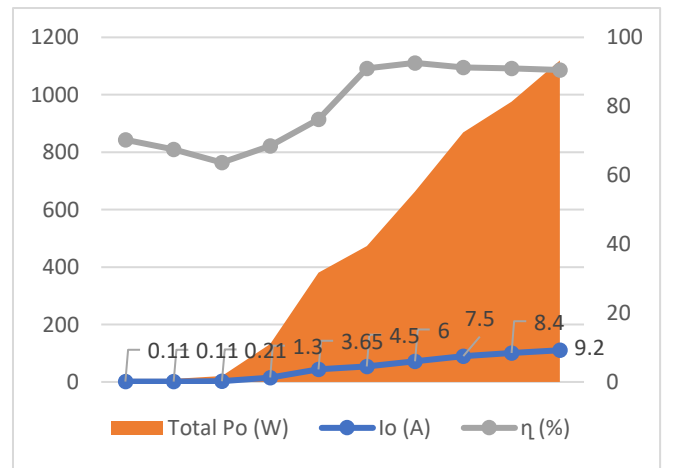


Fig 7. The graph shows the duty cycle versus current consumption (A) and efficiency ( $\eta$ ) of the DC/DC power converter.

In Fig.8, the temperature rise of MOSFET, ferrite core, copper used in the core, and battery temperature is plotted for power consumption. With the power above 1kW for a run

time of two hours, the MOSFET temperature rises over 60°C, whereas, other components operate at 50°C. The heatsink can dissipate this heat easily with the help of a dc fan.

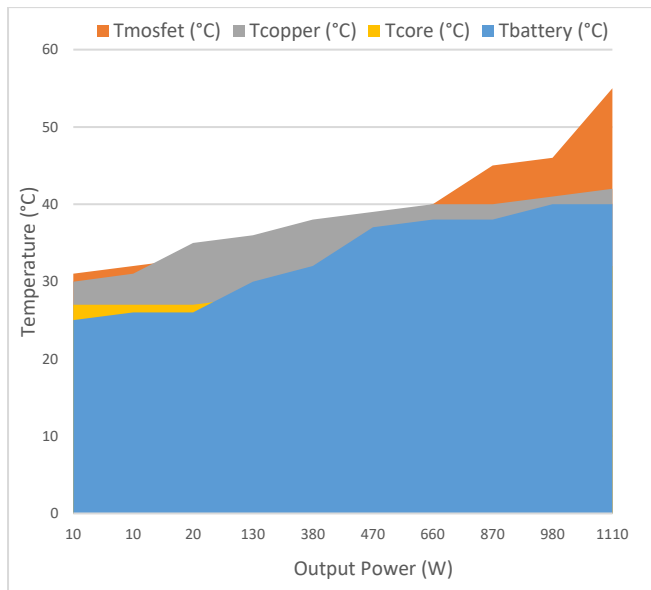


Fig 8. The graph shows the temperature spectrum of the various components with output power.

## V. CONCLUSION

In this paper, a detailed description of the design of an efficient, multiple output HF charger for the BMS integrated li-ion battery bank of the electric vehicle has been presented. For sophisticated and efficient controls, the discrete output is required by BMS with different voltage levels. With the HF multi-tapped transformer, the required output is carried out with 50V and a maximum of 10A current in each branch. The use of the ferrite core transformer in the proposed charger not only makes it much lighter but also more efficient. It has already been found that the proposed charger is almost 90% efficient. The experimental findings suggest that if the proposed charger is produced on a large scale, it has the potential to completely provide a feasible solution for updated battery chargers with enhanced efficiencies.

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